$$\vec{u} = [5, -3, 7]\,, \vec{v} = [-1, 6, 2]$$

$$\begin{aligned}\vec{u} \times \vec{v} &= [-3(2) - 7(6), 7(-1) - 2(5), 30 - 3] \\ &= [-48, -17, 27] \\ &= \vec{c}\end{aligned}$$

$$\begin{aligned}\vec{u} \cdot \vec{c} &= 5(-48) + (-3)(-17) + 7(27) \\ &= 0\end{aligned}$$